

THE INCOME FLUCTUATION PROBLEM

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- With complete markets, heterogeneity washes out of the aggregates, or reappears only as shocks to a representative agent.
- The data reject full insurance: individual consumption moves with individual income.
- So we need models in which markets are **incomplete**. We can build them in two ways:
 - restrict the **instruments**: model the assets we observe (bonds, equity, ...);
 - restrict the **transactions**: model the frictions that limit insurance (private information, limited commitment, ...).
- The first route gives **exogenously** incomplete markets, the second **endogenously** incomplete markets. We follow the first.

We build the **standard incomplete markets model** from the bottom up:

1. **Decision problem**: a single household facing constant prices (**this lecture**).
2. **Equilibrium**: a stationary economy populated by many such households, with prices that clear markets.
3. **Non-stationarity**: transitional dynamics and aggregate uncertainty.

Environment

- Infinite horizon, a single non-storable good, and a stochastic endowment y_t .
- The only way to move resources over time is through a one-period risk-free bond, traded at a discount, at price $q < 1$, which the household takes as given.
- The utility function satisfies the Inada conditions, and income is bounded:
 $0 < y_{\min} \leq y_t \leq y_{\max}$.

Household's problem

$$\begin{array}{ll} \max_{\{c_t, a_{t+1}\}_{t \geq 0}} & \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t u(c_t) \right] \\ \text{subject to} & c_t + q a_{t+1} = y_t + a_t \quad \text{for all } t, \\ & a_{t+1} \geq \underline{a} \quad \text{for all } t, \\ & \text{given } a_0, y_0, \text{ and } \underline{a}. \end{array}$$

What is a solution?

- Income is random, so the household chooses **contingent plans**: what to consume and save after each possible history $y^t = (y_0, \dots, y_t)$.
- A **solution** is therefore a pair of *stochastic processes*, $\{c_t, a_{t+1}\}_{t=0}^{\infty}$, with c_t and a_{t+1} functions of y^t and a_0 given.
- What are the **long-run properties** of these processes?
 - Do consumption and assets settle into a well-defined distribution, or do they drift without bound?
- Why care? With many such households, the cross section of assets at any date mirrors the distribution of a_t . An equilibrium needs it to settle down.

We answer this question in three steps, from the simplest setting to the most general:

1. **Deterministic income**: no risk.
2. **Idiosyncratic uncertainty with quadratic utility**: risk, but no precautionary motive.
3. **Idiosyncratic uncertainty with general preferences**: risk and precaution.

Optimality conditions

- Attach multipliers $\beta^t \lambda_t$ to the budget constraint and $\beta^t q \mu_t$ to the borrowing limit. The first-order conditions are

$$u'(c_t) = \lambda_t, \quad \lambda_t = \frac{\beta}{q} \mathbb{E}_t[\lambda_{t+1}] + \mu_t.$$

- Combining them gives the **Euler equation**

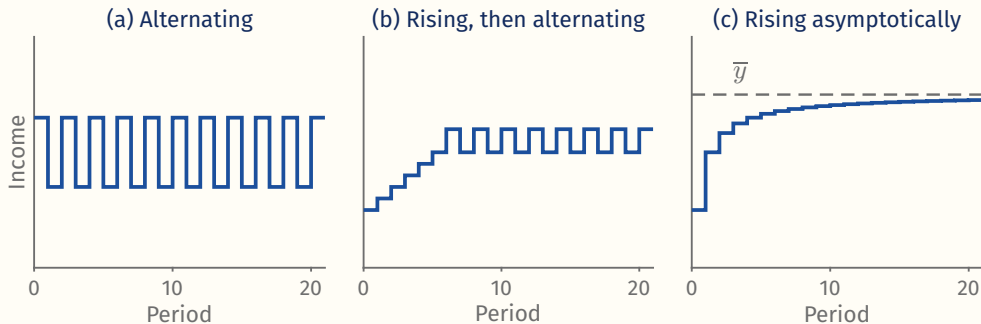
$$u'(c_t) = \frac{\beta}{q} \mathbb{E}_t[u'(c_{t+1})] + \mu_t, \quad \mu_t \geq 0, \quad \mu_t = 0 \text{ if } a_{t+1} > \underline{a}.$$

- Three motives to save meet in this condition: **intertemporal**, **consumption smoothing** and **precautionary**.
- The ratio β/q organizes the analysis: the higher it is, the stronger the intertemporal saving motive.

Part 1: deterministic income

Deterministic income

- Suppose first that income fluctuates, but in a way the household knows in advance. For example:



- A **solution** in this case is simply a pair of sequences, $\{c_t, a_{t+1}\}_{t=0}^{\infty}$.

The intertemporal budget

- Define x_t as the present value of current and future income:

$$x_t \equiv \sum_{s=0}^{\infty} q^s y_{t+s}.$$

- Recall that x_t is finite: income is bounded and $q < 1$.
- Iterate the budget constraint forward, and use $\lim_{T \rightarrow \infty} q^T a_{t+T} = 0$ at the optimum:

$$\sum_{s=0}^{\infty} q^s c_{t+s} = a_t + x_t \quad \text{for every } t.$$

- Why is that limit zero?

$\beta > q$: assets grow without bound

Claim: If $\beta > q$, then $\{c_t\} \rightarrow \infty$ and $\{a_t\} \rightarrow \infty$.

Proof.

- (1) Let $M_t \equiv (\beta/q)^t u'(c_t)$. The Euler equation implies that $\{M_t\}$ is decreasing.
- (2) $M_t \leq M_0$ implies in turn $u'(c_t) \leq (q/\beta)^t M_0 \rightarrow 0$, so $\{c_t\} \rightarrow \infty$.
- (3) For any K , eventually $c_t > K$, so $\sum_s q^s c_{t+s} > K/(1-q)$: the sum diverges.
- (4) Since $\sum_s q^s c_{t+s} = a_t + x_t$ and $\{x_t\}$ is bounded, $\{a_t\}$ must diverge too.

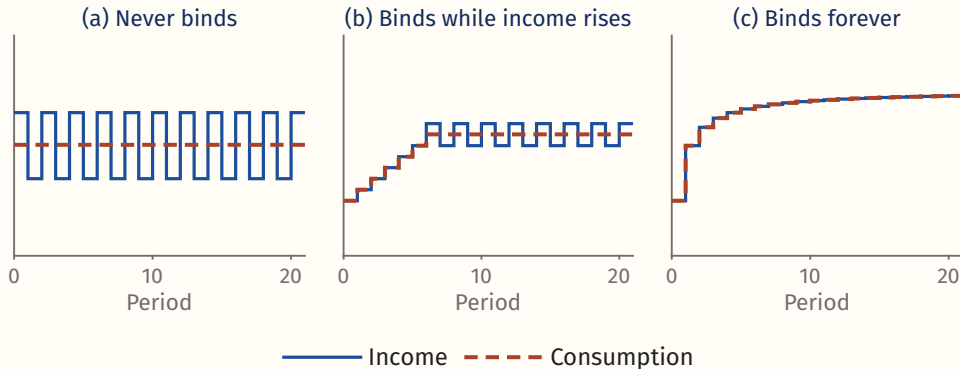


$\beta = q$: where consumption ends up

Claim: Let $\beta = q$. Consumption is nondecreasing, and if the borrowing limit:

- (i) never binds, then $c_t = (1 - q)(a_0 + x_0)$ for all t ;
 - (ii) binds for the last time at $T - 1$, then $c_t = (1 - q)(\underline{a} + x_T)$ for all $t \geq T$, where $x_T = \max_{t \geq 1} x_t$;
 - (iii) binds infinitely often, then $\{c_t\} \rightarrow (1 - q)(\underline{a} + \sup_{t \geq 1} x_t)$.
- In (i) and (ii), consumption is eventually constant: from then on, assets absorb all income fluctuations.
 - In (iii), the limit keeps binding, and assets cannot absorb them.

$\beta = q$: how often does the limit bind?



$\beta = q$: proof of the claim

Proof.

- (1) With $\beta = q$, the Euler equation is $u'(c_t) = u'(c_{t+1}) + \mu_t$: $\{c_t\}$ never falls, and is constant while the limit is slack.
- (2) *Case (i).* $\{c_t\}$ is constant at some c from $t = 0$, so $a_0 + x_0 = \sum_s q^s c_s = c/(1 - q)$. Then $c = (1 - q)(a_0 + x_0)$.
- (3) *Case (ii).*
 - $\{c_t\}$ is constant from T on and $a_T = \underline{a}$, so $c_T = (1 - q)(\underline{a} + x_T)$.
 - x_T is the peak: for $t \geq 1$, $\underline{a} + x_t \leq a_t + x_t = \sum_s q^s c_{t+s} \leq c_T/(1 - q) = \underline{a} + x_T$.
- (4) *Case (iii).* Bounding $\lim_t c_t$ from above and below gives the result (**PS2, Ex. 1**).



$\beta = q$, case (i): the permanent income hypothesis

- Case (i) illustrates Friedman's **permanent income hypothesis** (*Friedman, 1957*): consumption equals permanent income, the annuity value of total wealth, $c = (1 - q)(a_0 + x_0)$.

- Income fluctuations move wealth, not consumption. From the intertemporal budget,

$$a_t = \frac{c}{1 - q} - x_t \quad \implies \quad a_{t+1} - a_t = -(x_{t+1} - x_t).$$

- The household accumulates assets (saves) ahead of bad times, and depletes them or borrows ahead of good times.

$\beta < q$: consumption and assets stay bounded

Claim: Let $\beta < q$ and $\underline{a} > -\inf_t x_t$. Then:

- (i) the borrowing limit binds infinitely often;
- (ii) $\{c_t\}$ and $\{a_t\}$ are bounded.

- Consumption need not converge: with fluctuating income, it keeps moving, but within a band.
- With constant income, the band collapses to a point: once the limit binds, it binds forever, and $c_t = y + (1 - q) \underline{a}$ (**PS2, Ex. 2**).

$\beta < q$: proof of the claim

Proof.

- (1) While the limit is slack, $\mu_t = 0$ and $u'(c_{t+1}) = (q/\beta) u'(c_t) > u'(c_t)$: consumption **falls**.
- (2) *Part (i).* Suppose the limit never binds after some T . Then $\{u'(c_t)\}$ grows geometrically, so $\{c_t\} \rightarrow 0$ and $a_t + x_t = \sum_s q^s c_{t+s} \rightarrow 0$. But $a_t \geq \underline{a}$ requires $a_t + x_t \geq \underline{a} + \inf_t x_t > 0$.
- (3) *Part (ii).*
 - Consumption can rise only right after a binding date t , when $a_{t+1} = \underline{a}$, so $c_{t+1} \leq \sum_s q^s c_{t+1+s} = \underline{a} + x_{t+1} \leq \underline{a} + \sup_t x_t$. Before any binding date, $c_t \leq c_0 \leq a_0 + x_0$. So $\{c_t\}$ is bounded.
 - Then $a_t = \sum_s q^s c_{t+s} - x_t$ is bounded above, and $a_t \geq \underline{a}$.



Deterministic income: summary

	Borrowing limit	Long run
$\beta > q$	never or finitely often	$\{c_t\} \rightarrow \infty, \{a_t\} \rightarrow \infty$
$\beta = q$	never, finitely often, or forever	$\{c_t\}$ converges, $\{a_t\}$ bounded
$\beta < q$	binds infinitely often	$\{c_t\}, \{a_t\}$ bounded

- Recall that an equilibrium needs the cross section of assets to settle down. This rules out $\beta > q$.

Part 2: idiosyncratic uncertainty with quadratic utility

Why a quadratic benchmark?

- We want to isolate consumption smoothing under risk, so we switch off the other two motives:
 - $\beta = q$: the intertemporal motive is neutral;
 - quadratic utility, $u(c) = -\frac{1}{2}(\bar{c} - c)^2$ (this also drops Inada): marginal utility is linear, so $\mathbb{E}[u'(c)] = u'(\mathbb{E}[c])$ and risk leaves expected marginal utility unchanged;
 - no borrowing limit, only a no-Ponzi condition, $\lim_{T \rightarrow \infty} \mathbb{E}_t[q^T a_{t+T}] \geq 0$: the household can always borrow against bad news.
- The payoff is a closed form.
- Part 3 compares the general case with this benchmark at the same $\beta = q$: the difference is what precaution adds.

The consumption rule

- Marginal utility is linear and $\beta = q$, so the Euler equation becomes $\mathbb{E}_t[c_{t+1}] = c_t$: consumption is a **martingale** (Hall, 1978).
- As in Part 1, iterate the budget constraint forward, now taking expectations:

$$\sum_{s=0}^{\infty} q^s \mathbb{E}_t[c_{t+s}] = a_t + x_t, \quad x_t \equiv \mathbb{E}_t \left[\sum_{s=0}^{\infty} q^s y_{t+s} \right].$$

- The martingale property sets $\mathbb{E}_t[c_{t+s}] = c_t$ for all s , so

$$c_t = (1 - q) [a_t + x_t].$$

- This is Friedman's **permanent income hypothesis** under risk: the rule from Part 1, with x_t now an expectation.
- **Certainty equivalence**: this is the rule the household would follow if future income were known and equal to its conditional mean.

Consumption responds only to news

- Since $\mathbb{E}_{t-1}[c_t] = c_{t-1}$ and a_t is chosen at $t - 1$,

$$\Delta c_t = c_t - \mathbb{E}_{t-1}[c_t] = (1 - q) \sum_{s=0}^{\infty} q^s (\mathbb{E}_t - \mathbb{E}_{t-1}) y_{t+s}.$$

- Consumption changes only when the household **revises** the expected present value of its income, and it moves by the annuity value of that revision:
 - predictable income changes do not move consumption: $\mathbb{E}_{t-1}[\Delta c_t] = 0$;
 - how much a surprise moves consumption depends on how long it lasts: its **persistence**.
- To go further, we need specific assumptions about $\{y_t\}$.

Transitory shocks

- Take i.i.d. income, $y_t = \mathbb{E}[y] + \varepsilon_t$, so that $x_t = y_t + q \mathbb{E}[y]/(1 - q)$ and

$$\Delta c_t = (1 - q) \varepsilon_t.$$

- Consumption absorbs a transitory shock only by its **annuity value**, $1 - q = r/(1 + r) \approx r$, where r is the interest rate.
- Substituting the consumption rule into the budget constraint gives

$$a_{t+1} = a_t + \varepsilon_t.$$

- Consumption is a random walk**, and **so are assets**. Neither process has a limiting distribution: the cross-sectional variance grows linearly with t .
- Any borrowing limit would eventually bind with probability one, so we **could not have ignored it**.

Transitory and permanent shocks

- Add a permanent component: let income be the sum of a random walk and a transitory shock,

$$y_t = \eta_t + \varepsilon_t, \quad \eta_t = \eta_{t-1} + \nu_t,$$

with ε_t and ν_t i.i.d., mean zero and mutually independent. (Here we drop the assumption that income is bounded.)

- The revisions are $(\mathbb{E}_t - \mathbb{E}_{t-1}) y_{t+s} = \nu_t + \varepsilon_t \mathbf{1}\{s = 0\}$, so

$$\Delta c_t = \nu_t + (1 - q) \varepsilon_t.$$

- Consumption absorbs permanent shocks **one for one**, and **assets again follow** $a_{t+1} = a_t + \varepsilon_t$: permanent shocks never touch assets.

What we take from the quadratic case

- The bond is a good self-insurance device against **transitory** shocks and a poor one against **persistent** shocks. Other income processes: **PS2, Ex. 3**.
- Under smoothing alone, the household consumes only the annuity value of a transitory shock, $(1 - q) \varepsilon_t$, and keeps the rest in assets for good: assets have a **unit root**, $a_{t+1} = a_t + \varepsilon_t$. Smoothing determines how wealth moves, **not where it settles**.
- This already hints that, in a stationary general equilibrium, we will need $\beta < q$.
- Next: general preferences, where the borrowing limit can no longer be ignored.

Part 3: idiosyncratic uncertainty with general preferences

Back to the general problem

- We return to the problem as stated at the outset: general preferences and a borrowing limit, now with $\beta = q$.
- We add one assumption: any finite run of $y_{\min}$ draws has positive probability.
- With $\beta = q$, the Euler equation reads

$$u'(c_t) = \mathbb{E}_t[u'(c_{t+1})] + \mu_t.$$

The natural borrowing limit

- The **natural borrowing limit** is $\underline{a}^n \equiv -y_{\min}/(1 - q)$: the largest debt that can be repaid **with certainty**, even if income stays at $y_{\min}$ forever.
- Why the household never borrows up to it: let $d_t \equiv a_t - \underline{a}^n$ be the **distance to the limit**. When income is $y_{\min}$, the budget constraint becomes

$$d_{t+1} = \frac{d_t - c_t}{q}.$$

- If $d_t \leq 0$, along a run of $y_{\min}$ draws any positive consumption makes d negative, and from then on it falls at rate $1/q$: $\{d_t\} \rightarrow -\infty$, breaching any finite limit. The household would have to consume zero throughout a run that has positive probability: this cannot be optimal.
- So the household keeps $a_{t+1} > \underline{a}^n$ on its own: the natural limit **never binds**.

Natural versus ad hoc limits

- Any $\underline{a} \leq \underline{a}^n$ is equivalent to $\underline{a}^n$: it never binds, and the Euler equation holds with **equality** at every date.
- A limit $\underline{a} > \underline{a}^n$ is **ad hoc**: along some income paths it binds, even under the optimal plan.
- For now we set $\underline{a} = \underline{a}^n$. With $\beta = q$ the Euler equation becomes

$$u'(c_t) = \mathbb{E}_t[u'(c_{t+1})].$$

$\beta = q$: marginal utility converges

- $\{u'(c_t)\}$ is a **nonnegative martingale**. By the martingale convergence theorem, it converges almost surely: along each path, $u'(c_t) \rightarrow M_\infty$, a finite limit that may differ across paths.
- On paths where $M_\infty > 0$, consumption converges to a finite level. On paths where $M_\infty = 0$, consumption grows without bound.

Claim: If income is **sufficiently stochastic**, $M_\infty = 0$ almost surely, so $\{c_t\} \rightarrow \infty$ and $\{a_t\} \rightarrow \infty$ (Chamberlain and Wilson, 2000).

- The same holds with an ad hoc limit: marginal utility is then a nonnegative **supermartingale**, which still converges.

$\beta = q$: why consumption cannot settle (intuition)

- Suppose consumption settled at some c_∞ . With i.i.d. income, the budget constraint gives $a_{t+1} = (a_t + y_t - c_\infty)/q$.
- Let $a^\star \equiv (c_\infty - \mathbb{E}[y])/(1 - q)$. Deviations from it compound at rate $1/q$:

$$a_{t+1} - a^\star = \frac{(a_t - a^\star) + (y_t - \mathbb{E}[y])}{q}.$$

- Assets stay bounded only if future shocks exactly offset today's gap: a number known today. With random income, this has probability zero. The gap grows instead:
 - toward $-\infty$: assets breach the borrowing limit;
 - toward $+\infty$: wealth grows without bound while consumption stays at c_∞ , which cannot be optimal.
- So, with risk, consumption cannot settle. Without risk, it can: in Part 1, $\{c_t\}$ converges.

Precautionary saving

- Compare with Part 2 at the same $\beta = q$: there assets wander; here they **diverge**. The difference is precaution:
 - a long run of bad draws would force consumption ever lower;
 - households save to guard against that contingency;
 - with $\beta = q$, nothing offsets this motive, and wealth diverges.
- The precautionary saving motive can emerge from two sources: the shape of preferences or the borrowing limit.

Source 1: prudence

- Two periods and **no borrowing limit**: $\max u(c_0) + \beta \mathbb{E}[u(c_1)]$ subject to $c_0 + q a_1 = w$ and $c_1 = a_1 + y_1$.

- The optimal a_1 solves

$$q u'(w - q a_1) = \beta \mathbb{E}[u'(a_1 + y_1)].$$

- A mean-preserving spread in y_1 raises the right-hand side if and only if u' is convex. Then a_1 rises.
- A positive third derivative is called **prudence** (Kimball, 1990): risk raises the expected marginal value of wealth, and hence saving.
- With quadratic utility, $u''' = 0$: no prudence, and certainty equivalence.

Source 2: borrowing limits

- Prudence is **not necessary**. With quadratic utility and an ad hoc limit, the chance that the limit binds in the future already raises saving today (*Carroll and Kimball, 2001*).
- A limit that may bind makes next period's consumption function **concave** in wealth. Bad draws cut consumption by more than good draws raise it, so expected marginal utility rises even though u' is linear.
- Both sources are present with CRRA utility and a borrowing limit, the workhorse case in quantitative work.

$\beta < q$: impatience restores stationarity

- With $\beta = q$, wealth diverges. We now show that if the household is impatient, $\beta < q$, consumption and assets **stay bounded**.
- Intuition: impatience pulls wealth down; precaution pushes it up, but weakens as wealth grows, so eventually impatience wins.
- To show it, we write the household's problem in **recursive form**, with i.i.d. income. The same formulation is the starting point for the **numerical** methods of the next lecture.

The Bellman equation

- The state is (a, y) :

$$V(a, y) = \max_{c, a'} u(c) + \beta \mathbb{E}[V(a', y')] \quad \text{s.t.} \quad c + q a' \leq y + a, \quad a' \geq \underline{a}.$$

- With i.i.d. income, y does not help forecast y' , so a and y matter only through their sum, **cash on hand** $m \equiv a + y$:

$$V(m) = \max_c u(c) + \beta \mathbb{E}[V(q^{-1}(m - c) + y')] \quad \text{s.t.} \quad c \leq m - q \underline{a}.$$

- One state variable instead of two.

Optimality in recursive form

- Let $c^*(m)$ be the policy function, $m' = q^{-1}(m - c^*(m)) + y'$, and $\mu \geq 0$ the multiplier on $c \leq m - q\underline{a}$, the borrowing limit.
- The first-order and envelope conditions are

$$u'(c^*(m)) = \frac{\beta}{q} \mathbb{E}[V'(m')] + \mu, \quad V'(m) = \frac{\beta}{q} \mathbb{E}[V'(m')] + \mu.$$

- Hence $V'(m) = u'(c^*(m))$, and we recover the Euler equation

$$u'(c^*(m)) = \frac{\beta}{q} \mathbb{E}[u'(c^*(m'))] + \mu.$$

- If $\mu > 0$, the **borrowing limit binds**: $c^*(m) = m - q\underline{a}$.

Consumption rises with cash on hand

Claim: $c^*(m)$ is strictly increasing in m , and saving, $s(m) \equiv m - c^*(m)$, is non-decreasing in m .

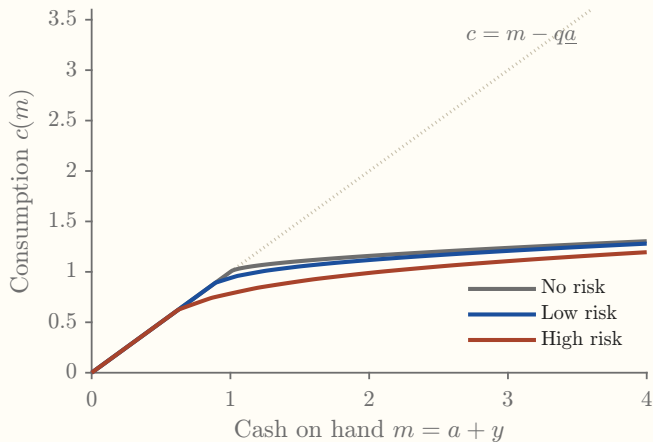
Proof.

- (1) Under standard assumptions, V is increasing and strictly concave; assume also that it is differentiable.
- (2) Take $m > \tilde{m}$. Strict concavity gives $V'(m) < V'(\tilde{m})$.
- (3) By the envelope condition, $u'(c^*(m)) < u'(c^*(\tilde{m}))$.
- (4) Since u' is decreasing, $c^*(m) > c^*(\tilde{m})$.
- (5) *Saving.* Suppose $s(m) < s(\tilde{m})$. Then the limit is slack at $\tilde{m}$, and since V' is decreasing,

$$u'(c^*(\tilde{m})) = \frac{\beta}{q} \mathbb{E}[V'(s(\tilde{m})/q + y')] < \frac{\beta}{q} \mathbb{E}[V'(s(m)/q + y')] \leq u'(c^*(m)),$$

so $c^*(\tilde{m}) > c^*(m)$, contradicting step (4).

The consumption function



l.i.d. income, $y \in \{0.8, 1.2\}$ (low risk) or $\{0.5, 1.5\}$ (high risk).

- CRRA with $\sigma = 2$, $\beta = 0.95$, $r = 3\%$, $\underline{a} = 0$.
- Below a threshold of m , the **borrowing limit binds**.
- Above it, c^* is concave: the marginal propensity to consume **falls with wealth**.
- More risk shifts c^* **down**: this gap is precautionary saving.

$\beta < q$: cash on hand is bounded

Claim: Let $\beta < q$, and suppose absolute risk aversion vanishes as consumption grows (e.g. CRRA). Then cash on hand is bounded (*Schechtman and Escudero, 1977*).

- Idea of the proof: find a level $\hat{m}$ above which cash on hand falls **whatever the income draw**. Cash on hand can then never rise above $\max\{m_0, \hat{m}\}$.

$\beta < q$: proof that cash on hand is bounded

Proof.

- (1) Let $\bar{m}(m) \equiv q^{-1}(m - c^*(m)) + y_{\max}$, the highest possible cash on hand next period. For large m the limit is slack; multiply and divide the Euler equation by $u'(c^*(\bar{m}(m)))$:

$$u'(c^*(m)) = \frac{\beta}{q} R(m) u'(c^*(\bar{m}(m))), \quad R(m) \equiv \frac{\mathbb{E}[u'(c^*(m'))]}{u'(c^*(\bar{m}(m)))} \geq 1.$$

- (2) **Suppose** $R(m) \rightarrow 1$ **as** $m \rightarrow \infty$ (next slide: this follows from the hypothesis). Since $\beta/q < 1$, there is an $\hat{m}$ such that, for all $m \geq \hat{m}$,

$$u'(c^*(m)) < u'(c^*(\bar{m}(m))) \implies c^*(m) > c^*(\bar{m}(m)) \implies \bar{m}(m) < m.$$

- (3) Above $\hat{m}$, cash on hand **falls for sure**. Below $\hat{m}$, it can rise, but not above $\hat{m}$: saving is nondecreasing, so $\bar{m}(m) \leq \bar{m}(\hat{m}) < \hat{m}$ for $m \leq \hat{m}$.
- (4) Hence $m_t \leq \max\{m_0, \hat{m}\}$ at every date: **a bound**.

When does $R(m)$ go to one?

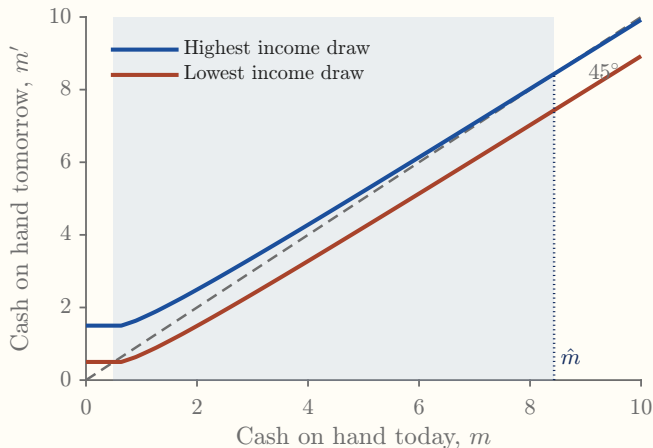
- Heuristically, linearize $u'(c^*(m'))$ around $\bar{m}$ and take expectations:

$$R(m) \approx 1 + \alpha(c^*(\bar{m})) c^{*'}(\bar{m}) (y_{\max} - \mathbb{E}[y]),$$

where $\alpha(c) = -u''(c)/u'(c)$ is the coefficient of **absolute risk aversion**.

- Since $0 < c^{*'} \leq 1$, a sufficient condition is $\alpha(c) \rightarrow 0$ as $c \rightarrow \infty$: absolute risk aversion **vanishes** with wealth.
- CRRA satisfies it: $\alpha(c) = \sigma/c$. CARA does not: its precautionary motive never fades.

$\beta < q$: the bound in a picture

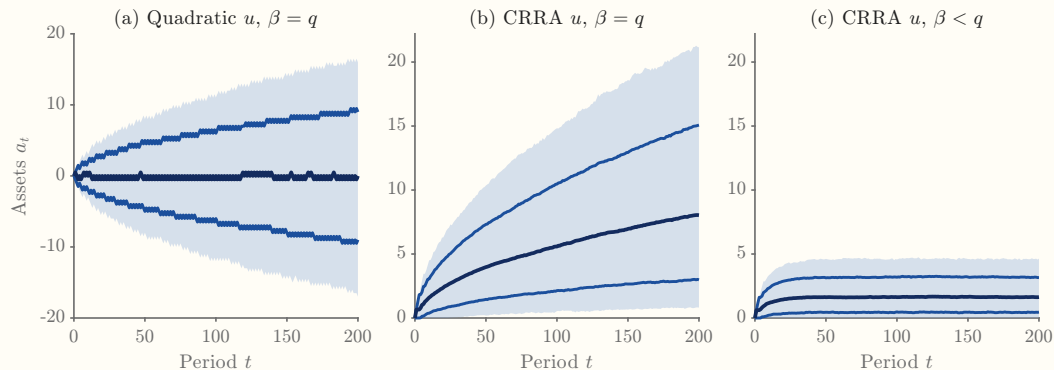


High-risk calibration, $r = 1\%$ (Deaton, 1991).

- Next period's cash on hand, $m' = s(m)/q + y'$, for the lowest and highest income draws.
- The highest draw crosses the 45° line at $\hat{m}$: beyond it, cash on hand falls whatever the draw.
- Starting in the shaded band, cash on hand **never leaves it**.

What boundedness means for a cross section

Recall: the cross section of assets at any date mirrors the distribution of a_t .



10,000 households, $a_0 = 0$, i.i.d. income $y \in \{0.5, 1.5\}$; panels (b)–(c) impose $a' \geq 0$. Lines: 10th, 50th and 90th percentiles; band: 1st to 99th.

Stochastic income: summary

Deterministic income	Stochastic income
$\beta > q$ $\{c_t\} \rightarrow \infty, \{a_t\} \rightarrow \infty$	$\{c_t\} \rightarrow \infty, \{a_t\} \rightarrow \infty$
$\beta = q$ $\{c_t\}$ converges, $\{a_t\}$ bounded	$\{c_t\} \rightarrow \infty, \{a_t\} \rightarrow \infty$
$\beta < q$ $\{c_t\}, \{a_t\}$ bounded	$\{c_t\}, \{a_t\}$ bounded, if $\alpha(c) \rightarrow 0$

- Uncertainty changes the answer exactly at $\beta = q$: **precautionary saving** pushes wealth to infinity.
- For a stationary distribution to exist, the household must be **strictly** less patient: $\beta < q$.

Toward general equilibrium

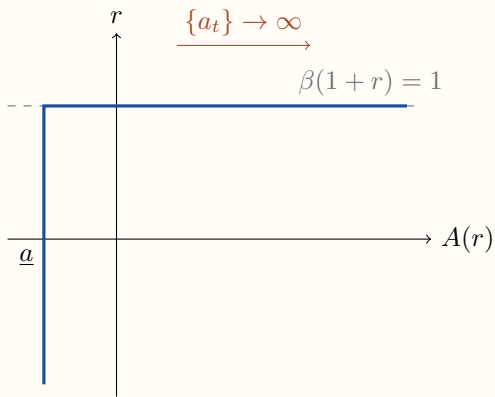
From one household to many

- Populate the economy with a continuum of these households. Income shocks are i.i.d. across households, so there is no aggregate risk.
- From now on, we index the bond price by the interest rate, $r = q^{-1} - 1$. Given r , every household's assets follow the same stochastic process. If its distribution converges in the long run to some λ_r , then, by a law of large numbers, λ_r is also the **cross section**.
- Average asset holdings define the long-run **supply of savings**:

$$A(r) \equiv \int a \, d\lambda_r(a).$$

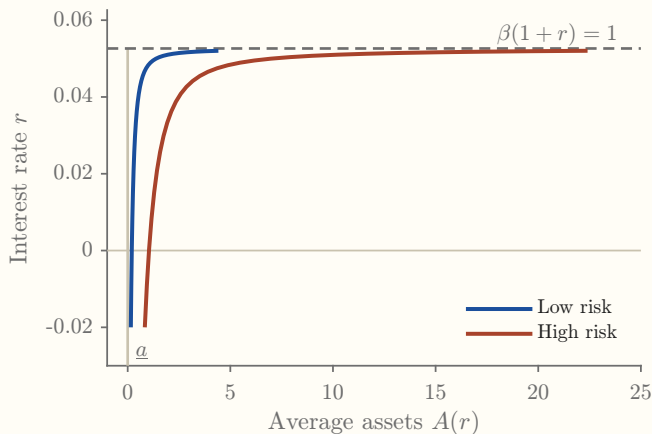
- $A(r)$ is well defined when individual assets are bounded. Our results tell us when that is the case.

Deterministic income: the supply of savings



- $\beta(1+r) < 1$: with constant income, everybody ends up at the limit, so $A(r) = \underline{a}$.
- $\beta(1+r) = 1$: any $A \geq \underline{a}$ is possible; where A lies depends on initial assets and on income paths.
- $\beta(1+r) > 1$: assets diverge; savings have no finite supply.
- The supply of savings is **perfectly elastic** at $r = (1 - \beta)/\beta$.

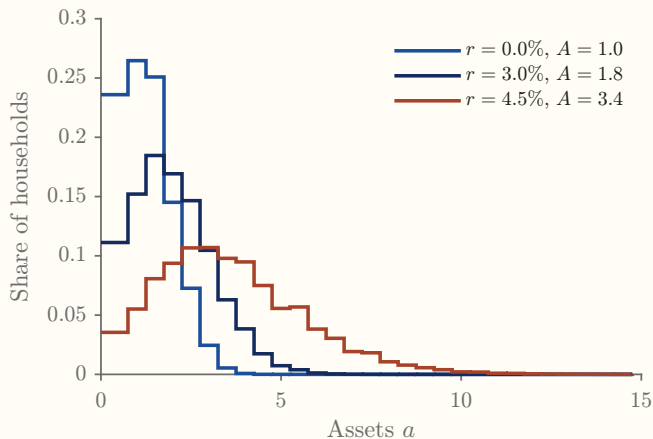
Stochastic income: the supply of savings



- $A(r) \rightarrow \infty$ as r approaches $(1 - \beta)/\beta$ from below.
- More income risk shifts $A(r)$ to the **right**: households save more at every r .

Same calibration as before; $\underline{a} = 0$. With an ad hoc limit, the problem is well defined for $r \leq 0$ too, i.e. $q \geq 1$.

The stationary distribution of assets



High-risk calibration.

- Each value of r gives a whole distribution, not just a mean.
- A mass of households sits **at the borrowing limit**.
- A higher r shifts mass to the right and stretches the **right tail**.

Closing the model

- Equilibrium requires a **demand** for savings to meet $A(r)$:
 - bonds in zero net supply among households: $A(r) = 0$ (*Huggett, 1993*);
 - capital demanded by firms: $A(r) = K(r)$ (*Aiyagari, 1994*).
- In either case, the equilibrium interest rate satisfies $\beta(1 + r^*) < 1$: precautionary saving **depresses** the interest rate below the complete-markets value.
- The equilibrium also delivers a **wealth distribution**, and it now matters for aggregates.
- Next lecture: solving the income fluctuation problem numerically.

Summary

- With a single risk-free bond, households **self-insure** by accumulating and running down assets.
- With quadratic utility, consumption is a martingale, responds only to news, and depends only on the mean of income: **certainty equivalence**.
- With prudence or borrowing limits, risk raises saving: **precautionary saving**.
- When $\beta = q$, that motive makes wealth diverge. When $\beta < q$ and absolute risk aversion vanishes with wealth, assets are **bounded**.
- Bounded assets give a well-defined supply of savings $A(r)$, the building block of the general equilibrium.

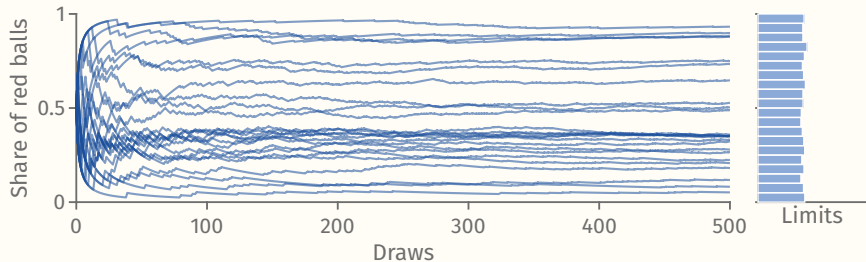
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Appendix: convergence to a random variable

An urn starts with one red and one blue ball. Each period, a ball is drawn at random and returned with another of the same color. The share of red balls is a martingale.

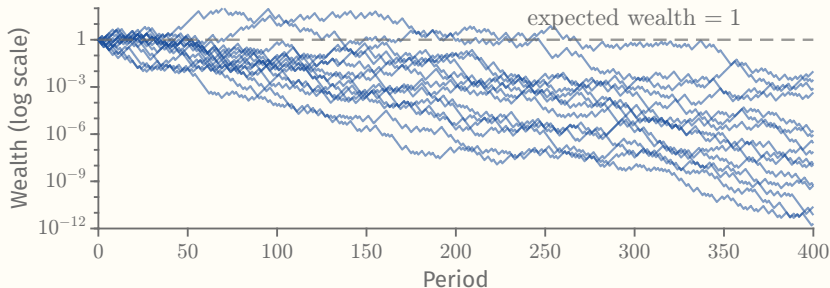
Readings: a worker whose employment raises the chance of future employment, and whose unemployment lowers it (**state dependence**); an economist whose every acceptance raises the odds of the next.



Each path settles at its own level: the limit is random, here uniform on $[0, 1]$. Early luck has permanent effects.

Appendix: convergence to zero

An investor reinvests all wealth each period in a bet that returns $+30\%$ or -30% with equal probability. The bet is fair, so expected wealth stays at its initial level: wealth is a martingale.



Yet the typical gross return is $\sqrt{1.3 \times 0.7} \approx 0.95 < 1$: log wealth drifts down, and wealth goes to zero on almost every path. The mean survives only through a vanishing fraction of very lucky paths. The same happens to $u'(c_t)$.

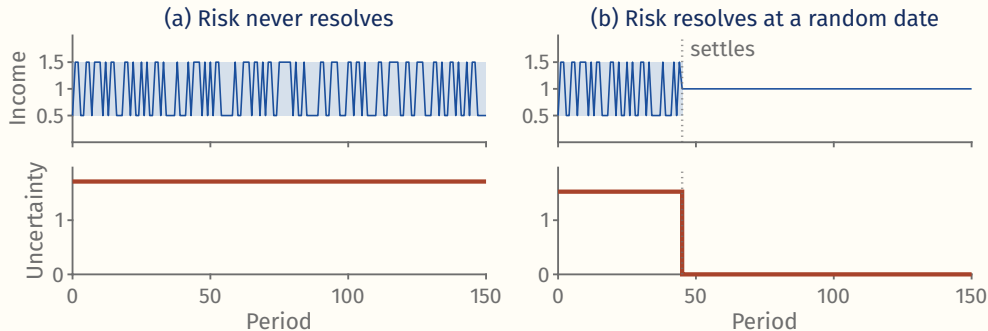
Appendix: what “sufficiently stochastic” means

- In simple terms: **risk never resolves**. However much time passes, the household stays uncertain about what its future income is worth.
- Precisely, with bounded income and $\beta R = 1$ (*Chamberlain and Wilson, 2000*): there is an $\varepsilon > 0$ such that, at every date t and after every history,

$$\text{Var}_t\left(\sum_{s \geq 0} q^s y_{t+s}\right) \geq \varepsilon.$$

- Why it matters: if consumption settled, the budget constraint would pin down the present value of future income from today's information. The condition rules this out, as in the intuition slide.
- Most income processes satisfy it: i.i.d. income, or a stationary Markov chain whose shocks never stop. It fails when risk eventually resolves ([illustration](#)).

Appendix: sufficiently stochastic, an illustration



Same shocks in both panels, $y_t = 1 \pm 0.5$. (a) Shocks never stop: uncertainty about future income stays constant, and the condition holds. (b) Each period, with a small probability, income settles at 1 forever (tenure, a fixed pension). The rules never change, but on every path risk eventually ends: the condition fails, and from then on consumption converges, as in Part 1.